

BRAINWARE UNIVERSITY
SCHOOL OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
Bachelor of Technology in Computer Science & Engineering – 2025
SEMESTER–III

Sl. No.	Course Code	Course Name	L	T	P	Evaluation Scheme		Total Marks	Credits	Course Type
						CIA	TEE			
1	BBS00014	Probability and Statistics	3	0	0	40	60	100	3	BSC
2	BES00007	Python Programming	3	1	0	40	60	100	4	ESC
3	BTS30101	Computer Organization and Architecture	3	1	0	40	60	100	4	PCC
4	BTS30102	Database Management Systems	3	1	0	40	60	100	4	PCC
5	BES09013	Python Programming Lab	0	0	2	40	60	100	1	ESC
6	BTS39103	Computer Organization and Architecture Lab	0	0	2	40	60	100	1	PCC
7	BTS39104	Database Management Systems Lab	0	0	2	40	60	100	1	PCC
8	BTS38101	Mini Project - I	0	0	0	40	60	100	2	PROJ
9	BAU00001	Constitution of India*	2*	0	0	0	0	0	0	AUC*
TOTAL			12+2*	3	6	320	480	800	20	

#MOOCs to earn additional credits for optional Honours/Minor Degree



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Course Code: BBS00014
Course Name: Probability and Statistics
Weekly Contact Hours: 3L
Credit(s): 3
Total Allotted Hours: 45H

Course Objective:

The objective of this course is to familiarize the students with statistical techniques. It aims to equip the students with standard concepts and tools at an intermediate to advanced level that will serve them well Towards tackling various problems in the discipline.

Pre-requisite(s): Basic Knowledge of Set Theory, Linear Algebra and Calculus.

Course Outcome: After the completion of the course, students would be able to:

CO1: Remember and understand the basic concepts of descriptive statistics and solve related problems, evaluate the problems of Correlation and Regression methods.

CO2: Recall the basic knowledge on fundamental probability concepts, including conditional probability and Bayes' Theorem, apply them to evaluate various related problems.

CO3: Understand the basic concepts of random variable, analyze several well-known distributions and solve related problems.

CO4: Remember the basic concepts of Estimation theory and Illustrate the concept of estimation techniques, analyze and evaluate different problems regarding estimation.

CO5: Describe and solve different methods of hypothesis testing and evaluate the results obtained in Z-test, T-test and F-test.

Module I: Descriptive Statistics

[9H]

Frequency Distribution, Measures of Central Tendency and Dispersion, Moments, Skewness and Kurtosis. Analysis of bivariate data, correlation, regression, method of curve fitting, rank correlation, multiple regression.

Module II: Probability Theory

[5H]

Review of Basic Probability, Independent events, Conditional Probability, Bayes' Theorem, Application of Bayes' Theorem.

Module III: Probability Distributions

[11H]

Random variables, Probability Distribution of Discrete and Continuous random variables, Joint Distributions, Expectation, Mean and Variance of Discrete, Continuous and Joint Distributions, Standard Distributions - Binomial, Geometric, Poisson, Uniform, Normal and Exponential distributions, Moments and Moment generating functions of Discrete and Continuous distributions.

Module IV: Parametric Estimation

[11H]

Population and Sample, sampling distributions, Estimation: Meaning and types, Maximum Likelihood estimator, Method of moment estimators, Unbiasedness, sufficiency, consistency, efficiency. Confidence Intervals

Module V: Testing of Hypothesis

[9H]

Statistical hypothesis, testing a hypothesis and significance, testing of hypothesis for Means and Variances, One-way ANOVA.



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Course Outcomes (Cos) and Modules Mapping:

Sl	Course Outcome (Cos)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1	CO1	M1	K1, K2, K3
2	CO2	M2	K1, K3, K5
3	CO3	M3	K2, K4
4	CO4	M4	K1, K2, K4, K5
5	CO5	M5	K2, K3, K5

Text books:

1. **"Fundamentals of Mathematical Statistics"**, S. C. Gupta, V. K. Kapoor, Sultan Chand and Sons, 12th Edition
2. **"Probability and Statistics for Engineers and Scientists"**, R. E. Walpole, R. H. Myers, S. L. Myers and K. Ye, Pearson Education, 8th Edition.
3. **"Introduction to Probability and Statistics for Engineers and Scientists"**, Sheldon M. Ross, Academic Foundation, 4th Edition.

Reference books:

1. **"Applied Statistics and Probability for Engineers"**, Douglas C. Montgomery, Wiley India, 5th Edition.
2. **"Probability & Statistics"**, Spiegel, M. R., Schiller, J. and Srinivasan, R. A. Tata McGraw Hill, 3rd Edition.



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Course Code: BES00007

Course Name: Python Programming

Weekly Contact Hours: 3L + 1T

Credit(s): 4

Total Allotted Hours: 60 H (45L + 15T)

Course Objective:

The objective of this course is to describe the core syntax and semantics of Python programming language and discover the need for working with the strings and functions. It also highlights the process of structuring the data using lists, dictionaries, tuples and sets along with use of regular expressions and built-in functions to navigate the file system. It also covers Object-oriented Programming concepts in Python.

Pre-requisite(s): Basic knowledge of computer programming language.

Course Outcome: After the completion of the course, students would be able to:

CO1: Describe the fundamental Concept of Python syntax and semantics and analyze the use control flow statements of Python Programming.

CO2: Express and measure proficiency in the handling of strings and functions.

CO3: Define the methods to create and manipulate Python programs by utilizing the data structures like lists, dictionaries, tuples and sets and compare their performance in Python Programming.

CO4: Define and evaluate the commonly used operations involving file systems, Modules and Packages.

CO5: Describe the Object-Oriented Programming concepts as used in Python and apply the concept to solve different type of program.

Module I: Basics of Python, Conditional & Control Statements

[10H]

History, Features, Setting up path, Working with Python, Basic Syntax, Variable and Data Types, Operator. If, If- else, Nested if-else, Looping, For, While, Nested loops, Break, Continue, Pass.

Module II: String Manipulation

[6H]

Accessing Strings, Basic Operations, String slices, Function and Methods.

Module III: Lists, Tuple & Dictionaries

[12H]

Introduction, Accessing list, Operations, Working with lists, Function and Methods.
Introduction, accessing tuples, Operations, Working, Functions and Methods.
Introduction, accessing values in dictionaries, Working with dictionaries, Properties.

Module IV: Functions & Modules

[7H]

Defining a function, Calling a function, Types of functions, Function Arguments, Anonymous functions, Global and local variables.

Importing module, Math module, Random module, Packages, Composition, Input-Output Printing on screen, Reading data from keyboard, Opening and closing file, Reading and writing files, Functions.



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Module V: Exception Handling & Object-Oriented Programming

[10H]

Exception, Exception Handling, Except clause, Try finally clause, User Defined Exceptions.

Classes and Objects, Creating Classes in Python, Creating Objects in Python, The Constructor Method, Classes with Multiple Objects, Class Attributes versus Data Attributes, Encapsulation, Inheritance, The Polymorphism.

Course Outcomes (COs) and Modules Mapping:

Sl	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1	CO1	M1	K1,K2
2	CO2	M2	K2,K3
3	CO3	M3	K2,K3,K4
4	CO4	M4	K2,K4
5	CO5	M5	K1,K3,K4

Text books:

1. Core Python Programming, 3ed Covers fundamentals to advanced topics like OOPS, Exceptions, Data structures, Files, Threads, Net, R. Nageswara Rao, Dreamtech Press, 1 September 2021.
2. Gowrishankar S, Veena A, "Introduction to Python Programming", CRC Press/Taylor & Francis, 1st Edition, 2018.

Reference books:

1. Jake VanderPlas, "Python Data Science Handbook: Essential Tools for Working with Data", 1st Edition, O'Reilly Media, 2016.
2. Aurelien Geron, "Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems", 2nd Edition, O'Reilly Media, 2019.
3. Wesley J Chun, "Core Python Applications Programming", Pearson Education India, 3rd Edition, 2015.
4. Miguel Grinberg, "Flask Web Development: Developing Web Applications with Python", O'Reilly Media, 2nd Edition, 2018, Khanna Publishers.



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Course Code: BTS30101

Course Name: Computer Organization and Architecture

Weekly Contact Hours: 3L + 1T

Credit(s): 4

Total Allotted Hours: 60H (45L + 15T)

Course Objective:

The objective of the course is to cover the basic concepts of computers, architecture of computer systems and electronic computers. It also helps students to design a simple computer using hardwired and micro programmed control methods along with different types of addressing modes and memory organization. It also focuses on input-output organization and different types of serial communication techniques and pipelining architecture with instruction cycle.

Pre-requisite(s): Fundamental of computers.

Course Outcome: After the completion of the course, students would be able to:

CO1: Define and explain the functional blocks of a computer, pipeline & memory.

CO2: Identify and design data representation in digital computers.

CO3: Express the concept of memory system and design pipeline architecture.

CO4: Apply data representation, instructions & evaluate memory system application.

CO5: Analyse and illustrate the pipeline architecture, memory and building blocks for various domains.

Module I: Functional blocks of a computer

[9H]

CPU, memory, input-output subsystems, control unit. Instruction set architecture of a CPU – registers, instruction execution cycle, RTL interpretation of instructions, addressing modes, instruction set. Case study – instruction sets of some common CPUs.

Module II: Data representation

[9H]

Signed number representation, fixed and floating point representations, character representation. Computer arithmetic – integer addition and subtraction, ripple carry adder, carry look-ahead adder, multiplication – shift-and add, Booth multiplier, carry save multiplier, Division restoring and non-restoring techniques, floating point arithmetic.

Module III: Memory system design

[9H]

Semiconductor memory technologies, memory organization. Peripheral devices and their characteristics: Input-output subsystems, I/O device interface, I/O transfers – program controlled, interrupt driven and DMA, privileged and non-privileged instructions, software interrupts and exceptions. Programs and processes – role of interrupts in process state transitions, I/O device interfaces – SCII, USB.

Module IV: Pipelining

[9H]

Basic concepts of pipelining, throughput and speedup, pipeline hazards. Parallel Processors: Introduction to parallel processors, Concurrent access to memory and cache coherency.



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Module V: Memory organization

[9H]

Memory interleaving, concept of hierarchical memory organization, cache memory, cache size vs. block size, mapping functions, replacement algorithms, write policies.

Course Outcomes (COs) and Modules Mapping:

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1	CO1	M1	K1,K2
2	CO2	M2	K3,K5
3	CO3	M3	K2,K6
4	CO4	M4	K3,K5
5	CO5	M5	K3,K4

Text books:

1. Computer Organization and Design: The Hardware/Software Interface, David A. Patterson and John L. Hennessy, 5th edition, Elsevier.
2. Computer Organization, Carl Hamacher, 6th Edition by, McGraw Hill Higher Education.

Reference books:

1. Computer Organization and Architecture: Designing for Performance, William Stallings, John Dean, 10th edition, Pearson Education.



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Course Code: BTS30102

Course Name: Database Management Systems

Weekly Contact Hours: 3L + 1T

Credit(s): 4

Total Allotted Hours: 60H (45L + 15T)

Course Objective:

In the modern age data is most valuable part. To extract some information from a huge amount of data and store for near future to collect some knowledge also an important area of development. In this scenario data management plays a vital role. From this course learners can understand every point of data managements and also it can implement its day to day life.

Pre-requisite(s): Set theory concept of mathematics.

Course Outcome: After the completion of the course, students would be able to:

CO1: Define the various aspects of data definition, classify different data models and cite the example of entity relationship diagram.

CO2: Describe and compare the several relational query languages, query processing and query optimization using open source and commercial DBMS.

CO3: Define different storage structure like B-Tree, B+ Tree to analyze the storage policy.

CO4: Explain and Analyze the transaction process to interpret serializability of scheduling process and concurrency control policy.

CO5: Evaluate and formulate the security policy of a database in conjunction with the advancement of database design.

Module I: Database system architecture

[9H]

Data Abstraction, Data Independence, Data Definition Language (DDL), Data Manipulation Language (DML). Data models: Entity-relationship model, network model, relational and object oriented data models, integrity constraints, data manipulation operations.

Module II: Relational query languages

[15H]

Relational algebra, Tuple and domain relational calculus, SQL3, Open source and Commercial DBMS - MYSQL, ORACLE, DB2, SQL server. Relational database design: Domain and data dependency, Armstrong's axioms, Normal forms (1NF, 2NF, 3NF, BCNF, 4NF, 5NF, 6NF), Dependency reservation, Lossless design. Evaluation of relational algebra expressions, Query equivalence, Join strategies, Query optimization.

Module III: Storage strategies

[10H]

Indices, B-trees, hashing. Transaction processing: Concurrency control, ACID property, Serializability of scheduling, Locking and timestamp based schedulers, Database recovery.

Module IV: Database Security

[7H]

Authentication, Authorization and access control.



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Module V: Data Warehousing

[4H]

Introduction to Data warehousing and data mining, SQL injection.

Course Outcomes (COs) and Modules Mapping:

Sl	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1	CO1	M1	K1, K2
2	CO2	M2	K2, K4
3	CO3	M3	K1, K4
4	CO4	M4	K2, K4
5	CO5	M5	K5, K6

Text books:

1. "Database System Concepts", Henry F. Korth and Silberschatz Abraham, Mc.Graw Hill, 7th edition.
2. "Fundamentals of Database Systems", Ramez Elmasri, Shamkant B. Navathe, Addison Wesley Publishing Edition.

Reference books:

1. "Principles of Database Systems", Ullman JD., Galgottia Publication. 2nd edition, 1993.
2. "Introduction to Database Management", Date C. J., Pearson, 3rd edition.



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Course Code: BES09013
Course Name: Python Programming Lab
Weekly Contact Hours: 2P
Credit: 1
Total Allotted Hours: 30H

Course Objective:

The objective of this course is to solve problems using Python programming language with proper syntax and logic.

Pre-requisite(s): Basic knowledge of computer programming language.

Course Outcome: After the completion of the course, students would be able to:

- CO1:** Analyze the Concept of Python syntax and semantics and apply the knowledge to solve the flow control of Python Programming.
- CO2:** Analyse and implement the concept of strings and functions to solve the python programming.
- CO3:** Apply the concept of lists, tuples, dictionaries and evaluate Python Program.
- CO4:** Apply and evaluate the commonly used operations involving File systems, Modules and Packages.
- CO5:** Apply and demonstrate OOPs concepts to solve the python program.

Module I: Variable in a Data Type, Conditional Statements & Pattern

[8H]

A Basic Introduction to Python Installation.

Write a program to Reverse the two-digit number.

Write a program in Python that will check for the case of a character from the user and write a program accordingly

Write a program in Python to find the maximum between three numbers.

Write a program in Python to find the factorial of a number.

Write a program in Python to find the GCD of two numbers.

Write a program in Python to check whether a given number is a Prime number or a Composite number.

Write a program in Python to print following pattern

AAAA
BBB
CC
D



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Write a program in Python to print the following pattern

```
1
2 2 2
3 3 3 3
```

Module II: String

[4H]

Write a program in Python to reverse a string using a for loop.
Write a program in Python to remove the vowels from a string.
Write a program in Python to find the count of a given character.

Module III: List and Dictionary

[6H]

Write a program in python to remove the duplicate all values from the list
Write a program in python to find the second max value in the list.
Write a program in python to create a list of squares of the first 20 numbers.
Write a program in python to check whether a matrix is symmetric, skew-symmetric, or not.
Write a program in python to find the sum of row-wise elements in a matrix.
Write a program in python in which a list is provided with positive and negative numbers and create a tuple from the list by taking positive numbers only.
Enter a string in python and create a dictionary with alphabets as keys and occurrences as values
Write a program in python to enter names as keys and marks as values for students. print the names according to marks in descending order.
Write a program in python to make a dictionary with name as key and date of birth as value and encode a menu-driven program for insert, delete, update etc.

Module IV: Function Calling, Function Arguments, Local and Global Variable

[6H]

Write a program in python to check if the string has all the alphabets “the quick brown fox jumps over the lazy dog”.
Write a program in python to write a function to find the number of lowercase and uppercase characters in string.
Write a program in python to write a function to check if palindrome or not without using string.
Write a program to show the use of local and global variables and lambda functions.
Write a program in python to perform linear search in a list.
Write a program in python to perform bubble sort in a list.
Write a program in python sum of the positive numbers passing them as arguments.

Module V: Exception Handling, File Handling & Object Oriented Programming

[6H]

Write a program in python to find the square root of a number using math.
Write a program in python to search a value by index.
Write a program in python to print $1+1/2+1/3+.....+1/n$.
Write a program in python to open the text file.
Write a program in python to perform read-write operations on the file.
Write a program in python to copy the content of one file to another file.
Write a program in python to demonstrate inheritance create a class called cse student from class student that has the extra attribute of graduation year. Override the function print info to print the details of the derived class.
Write a program in python to demonstrate polymorphism and define a function to register objects of both class student and CSE student.



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Extended Teaching–Learning:

Extended teaching–learning is supported using AI-assisted tools to enhance understanding of Python syntax, program logic, data structures, function design, debugging, and object-oriented concepts through guided explanations, error diagnosis, and code visualization, while ensuring independent student implementation and ethical use of AI.

AI Tools Usage: ChatGPT, Jupyter Notebook (AI-assisted extensions), Google Colab, Gemini-assisted code analysis and debugging tools

Course Outcomes (COs) and Modules Mapping:

Sl	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K4)
1	CO1	M1	K3,K4
2	CO2	M2	K3,K4
3	CO3	M3	K3,K5
4	CO4	M4	K3,K5
5	CO5	M5	K3,K4

Text books:

1. Engineering Drawing, Bhatt N.D., Panchal V.M. & Ingle P.R., Charotar Publishing House, 2014
2. Engineering Drawing and Computer Graphics, Shah, M.B. & Rana B.C. Pearson Education, 2008
3. Engineering Graphics, Agrawal B. & Agrawal C. M. , TMH Publication, 2012
4. Text book on Engineering Drawing, Narayana, K.L. & P Kannaiah Scitech Publishers, 2008
5. CAD Software Theory and User Manuals.



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Course Code: BTS39103

Course Name: Computer Organization and Architecture Lab

Weekly Contact Hours: 2P

Credit: 1

Total Allotted Hours: 30H

Course Objective:

The objective of this course is to learn HDL and apply the logic to design a simple or combinational gate using HDL.

Pre-requisite(s): Fundamental of computers.

Course Outcome: After the completion of the course, students would be able to:

CO1: Define the HDL and apply it to real life problems.

CO2: Analyze and design solutions to critical problems using the knowledge of HDL and its various logic gates.

CO3: Classify and evaluate different types of memory systems and registers to solve critical computational and logical problems.

CO4: Compare and synthesize the concepts of data representation and memory systems to solve difficult and complex problems

CO5: Apply and demonstrate the concepts of CPU, ALU, and their building blocks across various domains to solve real-life problems.

Module I: HDL introduction

[6H]

Basic digital logic programming with HDL

Module II: Arithmetic Operations

[6H]

8-bit Addition, Multiplication, Division.

Module III: Register & Memory

[6H]

8-bit Register design, Memory unit design and perform memory operations.

Module IV: ALU Design

[6H]

8-bit simple ALU design

Module V: CPU Design and Interfacing

[6H]

8-bit simple CPU design, Interfacing of CPU and Memory.

Extended Teaching–Learning:

Students will be engaged in hands-on lab exercises using simulators, VLab, AI-assisted code analysis tools and AI tutors in debugging, visualising logic flows, and understanding data representation and memory operations.

AI Tools Usage: ChatGPT, Gemini AI, Replit for code suggestion and debugging



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Course Outcomes (COs) and Modules Mapping:

Sl	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1	CO1	M1	K1,K3
2	CO2	M1,M2	K4,K6
3	CO3	M2	K2,K5
4	CO4	M2,M3	K4,K6
5	CO5	M4,M5	K2,K3

Text books:

1. Computer Organization and Design: The Hardware/Software Interface, David A. Patterson and John L. Hennessy, 5th edition, Elsevier.
2. Computer Organization, Carl Hamacher, 6th Edition by, McGraw Hill Higher Education.

Reference books:

1. Computer Organization and Architecture: Designing for Performance ,William Stallings, John Dean, 10th edition, Pearson Education



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Course Code: BTS39104
Course Name: Database Management Systems Lab
Weekly Contact Hours: 2P
Credit: 1
Total Allotted Hours: 30H

Course Objective:

The objective of this course is to design database schema for a given application and apply normalization, acquire skills in using SQL commands for data definition and data manipulation. It also help to develop solutions for database applications using procedures, cursors and triggers.

Pre-requisite(s): NIL

Course Outcome: After the completion of the course, students would be able to:

CO1: Apply and analyze the basic concepts of Database Systems and Applications.

CO2: Prepare and demonstrate the basics of SQL and construct queries using SQL in database creation and interaction.

CO3: Evaluate and assess the appropriate models and algorithms to transform available data into valuable and actionable information.

CO4: Appraise and develop a commercial relational database system (Oracle, MySQL) by writing SQL using the system.

Module I: Fundamental of DBMS **[3H]**

Converting table from ER diagram.

Module II: Create Database **[6H]**

Creating Database, Creating a Table Specifying Relational Data Types, Specifying Constraints Creating Indices.

Module III: Handling Table **[9H]**

Table and Record Handling INSERT statement.

Using SELECT and INSERT together DELETE, UPDATE, TRUNCATE statements DROP, ALTER statements.

Retrieving Data from a Database The SELECT statement Using the WHERE clause Using Logical Operators in the WHERE clause

Using IN, BETWEEN, LIKE, ORDER BY, GROUP BY and HAVING Clause.

Aggregate Functions, Combining Tables Using JOINS Subqueries.

Correlated nested query.

Module IV: Create Virtual Table **[6H]**

Creating view and operations on view.

Module V: Understanding PL/SQL **[6H]**

PL/SQL, stored procedure, function, cursor, trigger.



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Extended Teaching–Learning: Extended teaching–learning is supported using AI-assisted tools to enhance understanding of database design, normalization, SQL query formulation, procedural database programming, debugging, and interpretation of query results, while ensuring independent student implementation and ethical use of AI.

AI Tools Usage: ChatGPT, AI-assisted SQL query analysis tools, database visualization tools, AI-supported debugging and query optimization platforms

Course Outcomes (COs) and Modules Mapping:

Sl	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom’s Taxonomy from K1 to K6)
1	CO1	M1	K3,K4
2	CO2	M1,M2	K2,K3
3	CO3	M3	K3,K5
4	CO4	M4,M5	K5,K6

Text book:

1. SQL, PL/SQL the Programming Language of Oracle”, Ivan Bayross, BPB publication, 4th Edition.



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Course Code: BTS38101
Course Name: Mini Project - I
Weekly Contact Hours: NA
Credit(s): 2
Total Allotted Hours: NA

Course Objective:

The objective of this course is to implement hands-on skills or practical knowledge acquired through theory into practice in a particular domain of computer science.

Course Outcomes:

- CO1:** Demonstrate basic programming skills and apply foundational knowledge of Computer Science to solve simple problems.
- CO2:** Identify real-life problems and formulate a structured problem statement suitable for a small-scale project.
- CO3:** Develop basic logic and algorithmic thinking to implement a simple working model or application.
- CO4:** Work effectively in small teams, participate in discussions, and practice ethical collaboration.
- CO5:** Present and document the project development process and outcome in a concise and structured manner.

Course Description:

The students will work on a project to develop solutions/applications. The project culminates with a comprehensive report for evaluation. Besides weekly monitoring of the progress of project work, the continuous assessments will be based on three reviews, followed by the term-end evaluation as per the project evaluation guidelines and departmental instructions.



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Course Code: BAU00001

Course Name: Constitution of India

Weekly Contact Hours: 2L

Credit(s): 0

Total Allotted Hours: 30H

Course Objective:

The objective of the course is to inculcate the foundational principles of the Constitution and examine the different aspects of the Constitution and constitutionalism in the context of the Indian Legal system. Moreover, the course aims to bring out the normative presuppositions of the Indian Constitution as revealed in the Indian Constitution and how it has been interpreted by the judiciary from time to time and familiarize the students with the concept and working of Indian federalism and the legislative and executive relationship between the centre and the states under the Constitution.

Pre-requisite(s): Fair knowledge of class X on civics.

Course Outcome: After the completion of the course, students would be able to:

CO1: Define and demonstrate foundational provisions, basic features and concepts of the Indian Constitution by acquiring disciplinary knowledge.

CO2: Interpret and develop legal literacy to identify legal rights and constitutional remedies with the help of traditional and modern tools.

CO3: Examine and compare the working of state and central governments in India to develop analytical skills and scope of employability in administrative services.

CO4: Analyze and appraise the role of local institutions and electoral processes to promote democratic and civic participation.

Module I: Introduction to the Constitution of India

[6H]

The Making of Indian Constitution, Salient Features of Indian Constitution and Preamble

Module II: Fundamental Rights and Directive Principles

[6H]

Fundamental Rights, Directive Principles of State Policy & its Relevance, Fundamental Duties.

Module III: Union Government

[6H]

President, Prime Minister and Council of Ministers, Parliament, Organization and Jurisdiction of Supreme Court of India.

Module IV: State Government

[6H]

Governor, Chief Minister, State Legislature, Organization and Jurisdiction of High Court of State

Module V: Local-Self Government and Electoral Process in India

[6H]

Powers and Functions of Municipalities and Panchayats, Universal Adult Suffrage and Role of Election Commission



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SEMESTER–III

Course Outcomes (COs) and Modules Mapping:

Sl	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1	CO1	M1, M2	K1, K2
2	CO2	M2, M3, M4	K2, K3
3	CO3	M3, M4	K3, K4, K5
4	CO4	M5	K4, K5

Text books:

1. Introduction to the Constitution of India, Durga Das Basu, Lexis Nexis, 21st edition, 2021
2. The Constitutional Law of India, Dr. J. N. Pandey, Central Law Agency, 57th Edition, 2020

Reference books:

1. Indian Constitutional Law, M P Jain, LexisNexis, 8th Edition.
2. Constitutional Development and National Movement in India, V.D.Mahajan, S.Chand and Company, 1st Edition, 2005.